Task 04: Traffic Accident Data Analysis Plan

1. Data Preparation and Feature Engineering

The analysis requires converting raw data columns into meaningful categories.

- **Temporal Features:**
    - **Time of Day:** Convert the precise accident time into a **categorical feature** (e.g., Morning Rush (6 AM–10 AM), Daytime (10 AM–4 PM), Evening Rush (4 PM–7 PM), Night (7 PM–6 AM)).
    - **Day/Week/Month:** Extract the **Day of the Week** and **Month** to analyze weekly and seasonal trends.
- **Environmental Features:** Clean and standardize categories for **Road_Surface_Condition**, **Weather_Conditions**, and **Light_Conditions** to ensure consistent grouping for analysis.

2. Identifying Patterns and Trends (Analysis)

A. Temporal Patterns

Focus	Expected Pattern
Time of Day	Highest accident counts typically occur during **Evening Rush Hour** (4 PM – 7 PM) due to high traffic volume and driver fatigue.
Day of the Week	**Fridays** and **Saturdays** often show the highest counts due to weekend travel and increased impaired driving incidents.
Seasonal	Accidents may spike in **winter months** due to hazardous conditions (e.g., ice, fog) and reduced daylight hours.

B. Environmental Patterns

- **Weather and Severity:** Analyze accident counts and **severity** during adverse conditions (Rain, Snow, Fog) compared to clear weather. A high ratio of severe/fatal accidents during poor weather indicates a strong causal link.
- **Road Conditions:** Check if conditions like **Wet/Damp** or **Snow/Ice** correlate with a disproportionately high rate of severe accidents due to reduced tire traction.
- **Light Conditions:** Accidents occurring in **Darkness with No Street Lights** often have higher severity due to poor visibility.

3. Visualization and Hotspot Identification

A. Pattern Visualization

Visualizations should clearly display the distribution of accidents across key variables:

- **Bar Charts:** Used to display the distribution of accident frequency across discrete categories, such as **Time of Day** and **Day of the Week**.
- **Stacked Bar Charts:** Used to show the distribution of **Accident Severity** (e.g., Fatal, Serious, Slight) within categories like **Weather Conditions** or **Road Surface Condition**. This clearly reveals *risk*.

B. Hotspot Identification (Geospatial Analysis)

1. **Plotting:** Use the **Latitude** and **Longitude** data to map the location of every accident.
2. **Density Map/Heatmap:** Create a **Heatmap** to show the concentration of accidents. Areas with the darkest coloring are the **accident hotspots**.
3. **Targeted Analysis:** Once a hotspot is identified, perform focused analysis on the **road characteristics** (e.g., presence of a junction, speed limit, road type) and **time/weather conditions** specific to that location to diagnose the root cause and recommend tailored interventions.